DEPARTMENT OF ENGINEERING
INFORMATION AND COMPUTER

SIGNALS AND SYSTEMS

TASK 2 ANALOG SIGNALS

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Laboratory Section: Section 1 Friday 9:00-11:00
Student Semester: 4th

EXERCISE 1

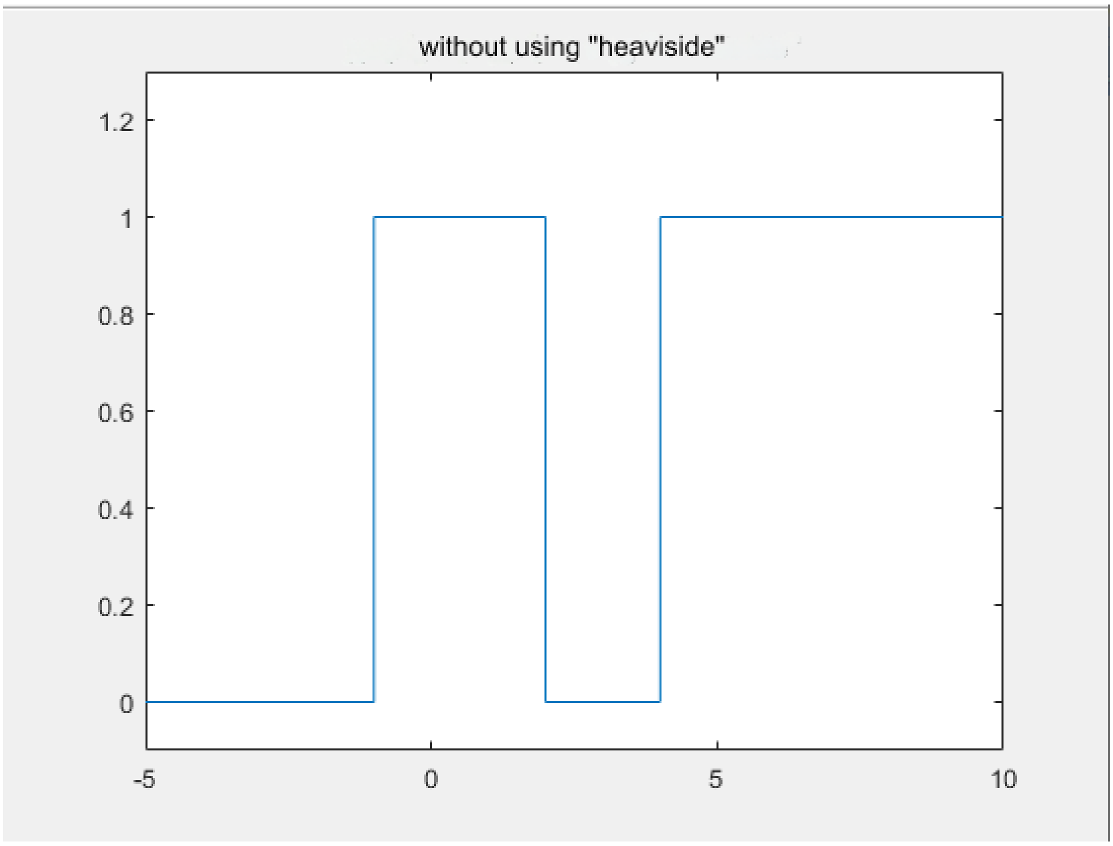
Commands:

a)
```
t 1 = -5:0.1:-1;
   t 2 = -1:0.1:2;
   t 3 = 2:0.1:4;
   t 4 = 4:0.1:10;
   u1 = zeros( size(t1));
u2 = ones( size(t2));
u3 = zeros( size(t3));
u4 = ones( size(t4));
t = [t1 t2 t3 t4];
u = [u1 u2 u3 u4];
   plot ( t , u )
title( 'without the use of "heaviside"' )
   ylim ( [-0.1,1.3]);
```

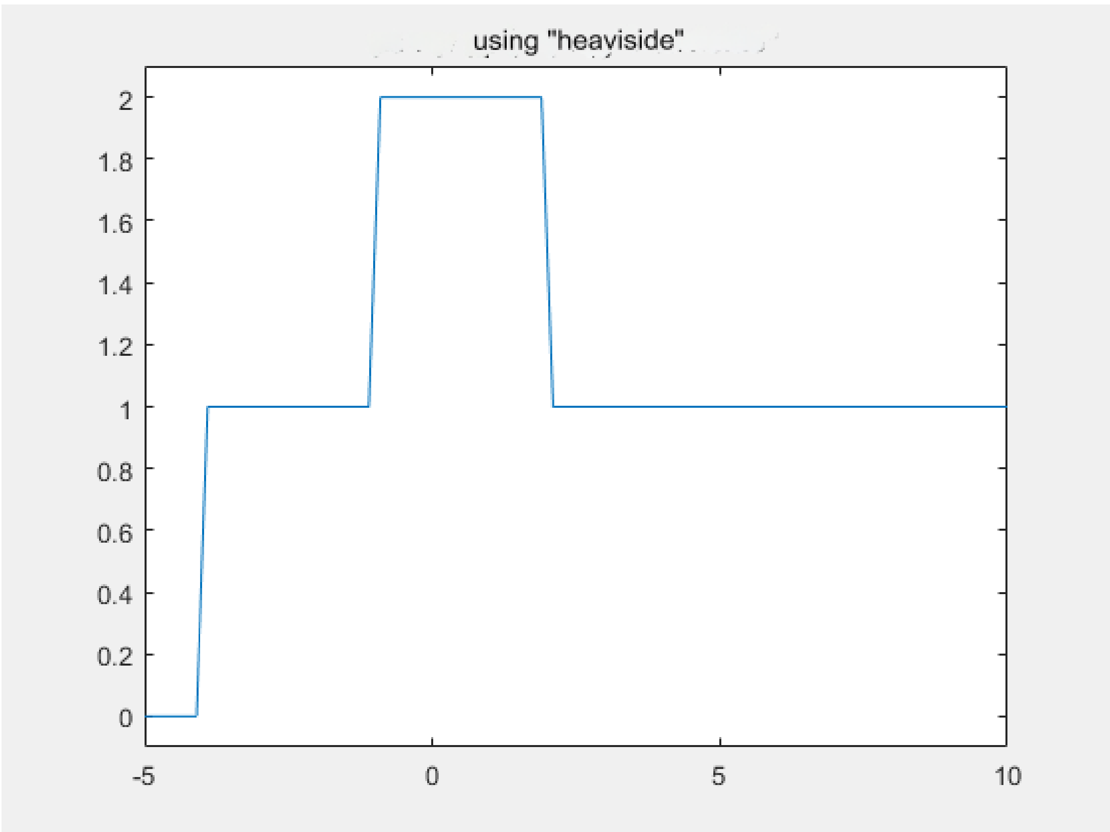
b)
```
t = -5:0.1:10;
   x1 = heavyside (t+1);
   x2 = heavyside (t-2);
   x3 = heavyside (t + 4);
x = x1 - x2 + x3;
   plot ( t , x )
title( 'using "heaviside"' )
   ylim([-0.1,2.1]);
```

Graphic representations:

a)


without using "heaviside"
1.2
1
0.8
0.6
0.4
0.2
0
-5
0
5
10

b)


using "heaviside"
2
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
-5
0
5
10

Documentation:

Exercise 1 asks us to plot the signal $x(t) = u(t+1) - u(t-2) + u(t-4)$ in two different ways: a) without using the " heaviside " function, b) using the " heaviside " function.

a)

For the first case, we use the method of drawing functions with many branches, in order to draw the above signal. In more detail, we divide the interval [-5,10] into 4 subintervals namely [-5,-1], [-1,2], [2,4], [4,10]. In "mathematical language" we find the sign of the polynomial by dividing the interval [-5,10], after first recording the roots which specifically for this polynomial are "-1,2,4". These subintervals are registered respectively in the variables " t 1", " t 2", " t 3" and " t 4". Immediately after, we make 4 vectors " u 1 ", " u 2 ", " u 3 " and " u 4 " in which we enter arrays of all zeros and arrays of all ones, thanks to the functions " zeros " and " ones " with parameters "(size (t v))" and dimensions equal to the arrays of subintervals " t 1 ", " t 2 ", " t 3 " and " t 4 ". Essentially, we plot the sign matrix of the polynomial, where the matrix of all zeros represents the negative sign in the given subinterval, while the matrix of all units represents the positive sign in the given subinterval. The subinterval arrays and vectors of all-zero and all-ones arrays are stored in the " t " and " u " vectors respectively. The plotting of the signal is done with the function " plot " and parameters the vectors "(t , u)". For a better visual representation of the graph, we used the function " ylim " with parameters "([-0.1,1.3])" so that the limit of the y- axis that will be displayed in the graph is from "-0.1 to 1.3". Finally, using the " title " function, for the sake of convenience we named the graph "without the use of " heaviside "".

b) For the second case we use the "

heaviside " function , where it does all the work we did in the first case. To begin with, we enter in a table " t " the interval of the signal "[-5,10]".

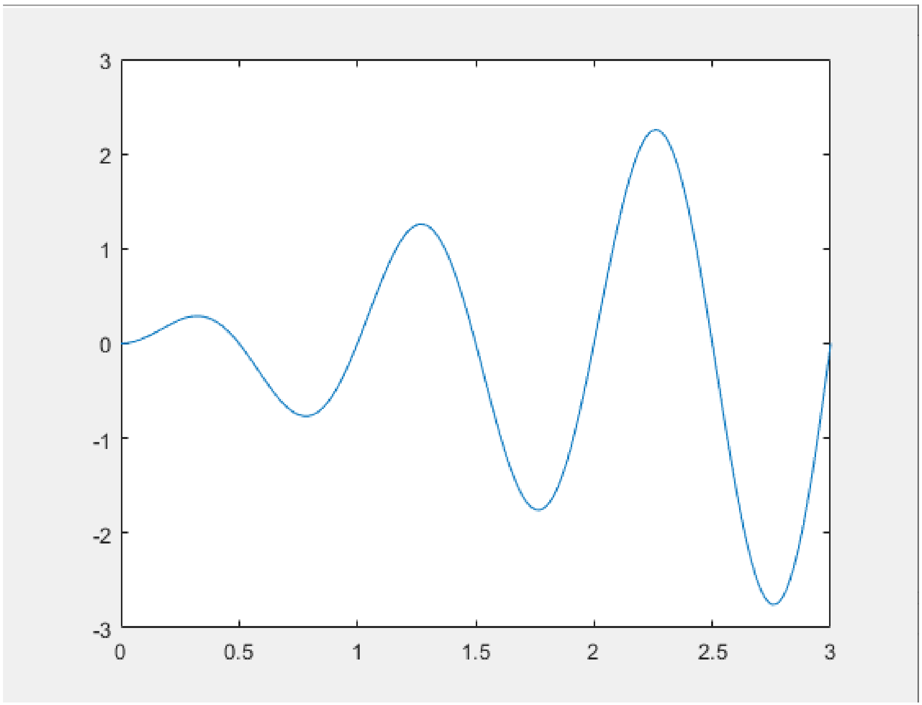
In three variables " x 1", " x 2" and " x 3" we define " u (t +1)", " u (t -2)" and " u (t -4)" respectively. We then assign the signal according to the definition to a variable " x ". The plotting of the signal is done with the " plot " function and "(t , x)" parameters . For a better visual representation of the graph, we used the function " ylim " with parameters "([-0.1,2.1])" so that the limit of the y- axis that will be displayed in the graph is from "-0.1 to 2.1". Finally, using the " title " function, for the sake of convenience we named the graph "using " heaviside "".

EXERCISE 2

Commands :

```
t = 0:0.01:3;
x1 = t.*sin(2*pi*t);
x2 = heaviside (t);
x3 = heavyside (t-3);
x = x1.*(x2-x3);
plot( t,x )
```

Graphic representations:



Documentation:

Exercise 2 asks us to plot the signal $x(t) = t\sin(2\pi t)(u(t) - u(t-3))$

First, we define the time interval " t " of the signal [0,3], as the roots are 0, ½, 1, 3. Next, we define the function " tsin (2π t)" in the variable " x 1", the function " u (t)" using " heaviside " in the variable " x 2" and the function " u (t -3)" using " heaviside " in the variable " x 3". Next, we define τ o signal " x (t)" in the variable " x ", as we defined the signals in the variables " x 1 ", " x 2 " and " x 3 ", { x 1(x 2- x 3)}. Finally, with the function " plot " and parameters the variables " t , x " we plot the signal.

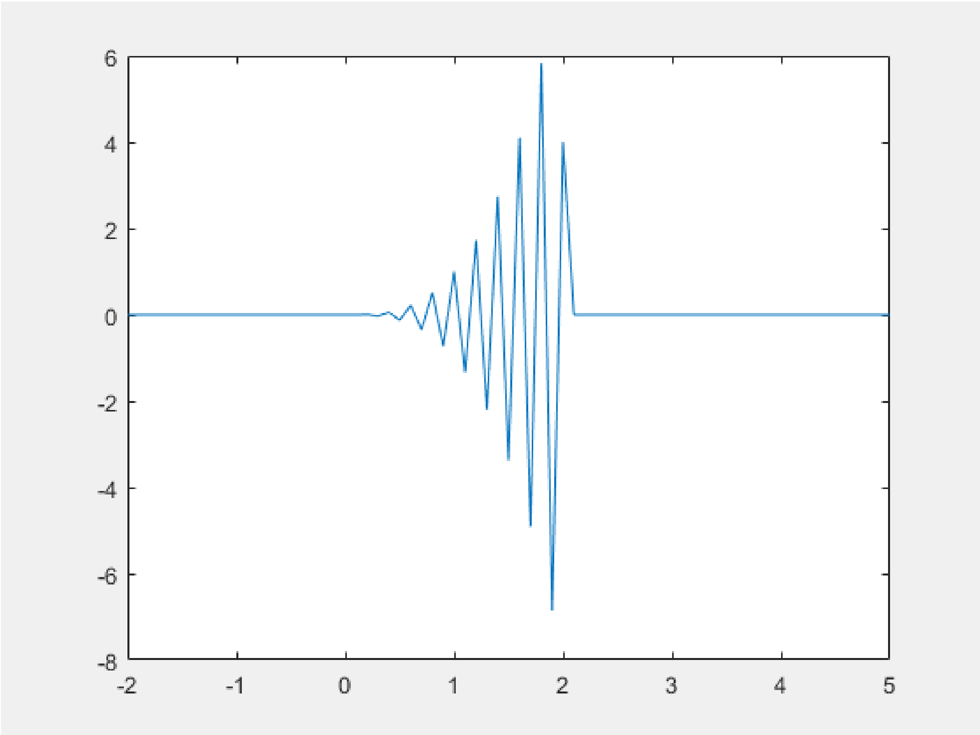
EXERCISE 3

Commands :

```
t = -2:0.1:5;
x1 = t.^3.* cos (10.*pi.*t);
x2 = heaviside (t);
x3 = heavyside (t-2);
```

```
s = x2 - x3;
x = x1.*s;
plot( t,x )
```

Plots shows :



Documentation:

Exercise 3 asks us to draw the signal $x(t) = t^3 \cos(10\pi t) p2(t-1)$, where pT (t), square pulse of duration T. Initially, the square pulse is defined as follows:

pT (t) = u (t + T /2) – u (t – T /2) =>
p 2(t - 1) = u (t – 1 + 2/2) – u (t – 1 – 2/2) =>
p 2(t - 1) = u (t) – u (t – 2)

We define the time interval " t ", according to the roots of the signal, the interval [-2,5]. Next, we define the signals " t .^3.* cos (10.* pi .* t)" in the variable " x 1", " u (t)" using the " heaviside " in " x 2" and " u (t - 2)" using " heaviside " on " x 3". Also , we define the square pulse duration T=2 in the variable " s " and then the final signal in the variable

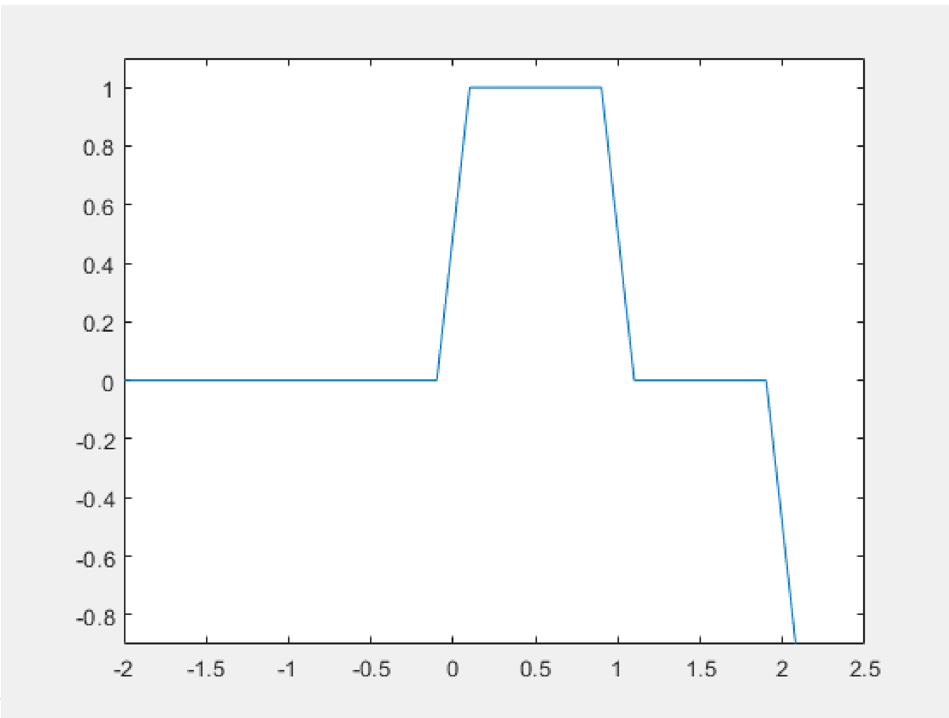
" x ". Finally, with the function " plot " and parameters the variables " t , x " we plot the signal.

EXERCISE 4

Commands :

```
t = -2:0.1:3;
x1 = heaviside (t);
x2 = heavyside (t-1);
x3 = heavyside (t-2);
x = x1 - x2 - x3;
plot( t,x )
ylim([-0.9,1.1]);
```

Graphic representations:



Documentation:

Exercise 4 shows us the above graph and asks us to express and plot the signal as a sum of ramp functions only. First, we define the time interval " t " of the signal [-2,3], according to the plot values on the x -axis . From the graph we find that the desired signal that the exercise asks us to express is " x (t) = u (t) – u (t -1) – u (t -2)". Next, we define the signal " u (t)" using the " heaviside " on the variable " x 1 ", the function " u (t -1)" using the " heaviside " on the variable " x 2 " and the function " u (t -2)" using the " heaviside " in the variable " x 3". Next, we assign the signal " x (t)" to the variable " x ", just as we assigned the signals to the variables " x 1 ", " x 2 " and " x 3 ", { x 1- x 2- x 3)} . Finally, with the function " plot " and parameters the variables " t , x " we plot the signal. For a better visualization of the signal we used the function " ylim " with parameters "([-0.9,1.1])".

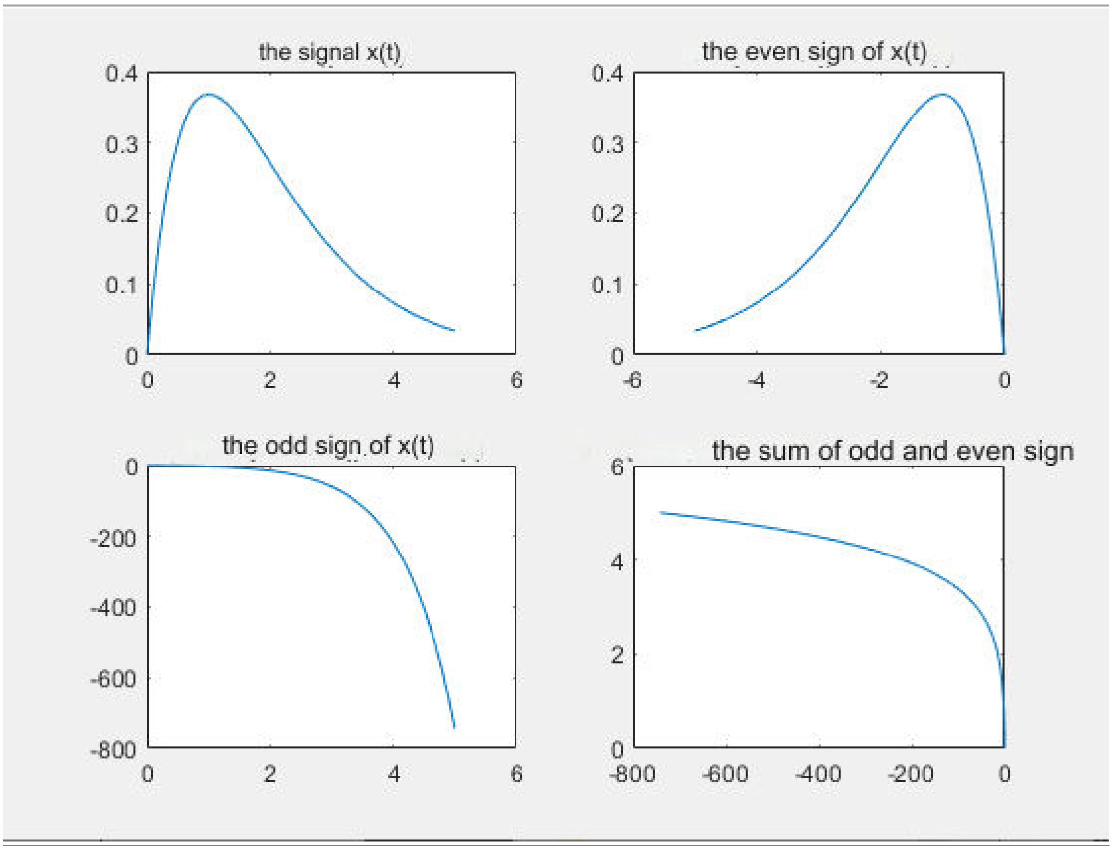
EXERCISE 5

Commands:

```
t1 = 0:0.1:5;
x1 = t.*exp(-t);
subplot( 2,2,1 );
plot( t1,x1 )
title( ' the signal x(t)' );
t2 = 0:-0.1:-5;
x2 = t.*exp(-t);
subplot( 2,2,2 );
plot(t2,x2)
title( 'the even sign of x(t)' );
```

```
x3 = -t.*exp(t);
subplot( 2,2,3 );
plot(t1,x3);
title( 'the odd signal of x(t)' );
s = x2 + x3;
subplot( 2,2,4 );
plot( s,t1 )
title( 'the sum of odd and even signs' );
```

Graphic representations:



Documentation:

Exercise 5 asks us to plot the sign $x(t) = te^{-t}, 0 \leq t \leq 5.$, the even sign, the odd and the sum of even and odd. We first define the intervals " t 1" and " t 2" the intervals [0,5] and [0,-5] respectively. Next, we assign the signal to the variable " x 1 " and with the function " plot " with parameters " t 1, x 1 " we plot the first signal in a 2 x 2 matrix thanks to the function " subplot ". We follow the same procedure for drawing the even signal with the difference that we define the signal in the variable " x 2", take as parameters in " plot " " x 2, t 2" and draw the second signal in a matrix of dimensions 2 x 2 thanks to the " subplot " function. For the odd signal we change the signs in the " t " variable, assign the signal to the " x 3" variable, take as parameters in " plot " " x 3, t 1" and plot the third signal in a 2 x 2 matrix thanks to in the " subplot " function. For the sum we define in the variable " s " the sum " x 2+ x 3" take as parameters in the " plot " " s , t 1" and draw the fourth signal in a matrix of dimensions 2 x 2 thanks to the function " subplot ". For convenience with the " title " function we set a title for each graph separately.

EXERCISE 6

Commands :

```
t = 0:0.1:5;
x = t.* cos ( 2*pi*t);
subplot( 2,2,1 );
plot( t,x )
title( 'x(t)' );
subplot( 2,2,2 );
plot( - t,x )
title( 'x(-t)' );
subplot( 2,2,3 );
plot( t./5,x)
title( 'x(t/5)' );
subplot( 2,2,4 );
plot( 1+3.* t,x )
title( 'x(1+3t)' );
subplot( 2,2,5 );
plot( -1-3.* t,x )
```

```
title ( ' x (-1-3 t )' );
```

<u>Graphic representations:</u>